

Nam Hee Kim

CURRICULUM VITAE

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Education

PhD - Embodied Intelligence • Computational Behavior

Helsinki, Finland

AALTO UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE

2021 - 2026

- Supervisors: Perttu Hämäläinen • Jaakko Lehtinen • Antti Oulasvirta
- Research interests: physics-based character animation • deep reinforcement learning • generative models
- Thesis (IN PROGRESS): Learning to Motion Plan—Advances in Algorithms, Architectures, and Applications
- NSERC Canada Graduate Scholarship-Doctoral (CGS-D) Award • Finnish Center for AI Open Application Award

MSc - Motion Control • Character Animation

Vancouver, Canada

UNIVERSITY OF BRITISH COLUMBIA, DEPARTMENT OF COMPUTER SCIENCE

2019 - 2021

- Supervisor: Michiel van de Panne
- Thesis: Directable Physics-based Character Animation
- NSERC Canada Graduate Scholarship-Master's (CGS-M) Award • Graduate Teaching Assistant Award

BSc - Bioinformatics • Data Science

Vancouver, Canada

UNIVERSITY OF BRITISH COLUMBIA, DEPARTMENT OF COMPUTER SCIENCE

2016 - 2018

- Ranked #1 in the Combined Major program cohort

BSc - Human-Computer Interaction • Bioinformatics

Houston, United States

RICE UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE, TRANSFERRED

2012 - 2016

- President's Honor Roll

Publications

10. **Kim, N. H.**, Liu, J. M., Lehtinen, J., Hämäläinen, P., O'Brien, J.F., & Peng, X.B. (2026). Robo-Saber: Generating and Simulating Virtual Reality Players. Computer Graphics Forum (Eurographics 2026).
9. Cheema, N., Xu, R., **Kim, N. H.**, Hämäläinen, P., Golyanik, V., Habermann, M., Theobalt, C. & Slusallek, P. (2023). Discovering Fatigued Movements for Virtual Character Animation. In ACM SIGGRAPH Asia 2023 Conference Proceedings (pp. 1-12).
8. Mensah, D., **Kim, N. H.**, Aittala, M., Laine, S. & Lehtinen, J. (2023). A Hybrid Generator Architecture for Controllable Face Synthesis. In ACM SIGGRAPH 2023 Conference Proceedings (pp. 1-10).
7. Cohan, S., **Kim, N.H.**, Rolnick, D. & van de Panne, M. (2022). Understanding the Evolution of Linear Regions in Deep Reinforcement Learning. In Proceedings of Neural Information Processing Systems (NeurIPS) 2022.
6. **Kim, N.H.**, Kirjonen, M. & Hämäläinen, P. (2022). Learning High-Risk High-Precision Motion Control. ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG) 2022.
5. Ng, B., Casazza, W., **Kim, N.H.**, Wang, C., Farhadi, F., Tasaki, S., Bennett, D., De Jager, P., Gaiteri, C. & Mostafavi, S.. (2021). Cascading Epigenomic Analysis for Identifying Disease Genes from the Regulatory Landscape of GWAS Variants. PLOS Genetics.
4. **Kim, N.H.**, Ling, H.Y., Xie, Z. & van de Panne, M.. (2021). Flexible Motion Optimization with Modulated Assistive Forces. In Proc. ACM Transactions on Graphics and Interactive Techniques (SCA 2021).

3. Xie, Z, Ling, H.Y., **Kim, N.H.**, & van de Panne, M.. (2020). ALLSTEPS: Curriculum-driven Learning of Stepping Stone Skills. In Proc. ACM SIGGRAPH / Eurographics Symposium on Computer Animation. **Best Paper Award**
2. **Kim, N.H.**, Xie, Z. & van de Panne, M.. (2020). Learning to Correspond Dynamical Systems. Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC), in Proc. Machine Learning Research (PMLR) 120:105-117. **Oral Presentation (Top 14 Papers of 131 Submissions)**
1. Ng, B., **Kim, N.H.**, Thirion, B., Poline, J., Greicius, M., Mostafavi, S., & IMAGEN Consortium. (2018). Associating IQ to Brain Networks with Manifold Regression. 2018 Organization for Human Brain Mapping Annual Meeting. (Peer-Reviewed by Abstract)

Selected Media Coverage

- Why Teach An AI To Climb Stepping Stones? - Two Minute Papers** 2021
<https://www.youtube.com/watch?v=ZZ-k0Rb8grA>
- Every Step They Took led to SCA Best Paper Award - University of British Columbia** 2021
<https://www.cs.ubc.ca/news/2021/01/every-step-they-took-led-sca-best-paper-award>
- Researchers Develop Hard-to-Hack Voting Machine - Voice of America** 2016
<https://www.voanews.com/usa/us-politics/researchers-develop-hard-hack-voting-machine>

Honors & Awards

- NSERC Canada Graduate Scholarship-Doctoral Program** 2022
 CAD 35,000/yr award to promote continued excellence in Canadian research; converted to PGS-D for study-abroad (CAD 21,000/yr)
- Finnish Center for AI (FCAI) Open Application Program-Doctoral Funding** 2021
 Full 4-year funding provided to exceptional doctoral students interested in tackling challenges in machine learning and in artificial intelligence
- UBC Department of Computer Science Graduate Teaching Assistant Award** 2021
 Awarded to graduate teaching assistants recommended by faculty and students for excellent teaching efforts.
- NSERC Canada Graduate Scholarship-Master's Program** 2020
 CAD 17,500 award for students who demonstrate a high standard of achievement in undergraduate and early graduate studies
- UBC Department of Computer Science Academic Award of Excellence (Combined Majors)** 2019
 Awarded to the individual graduating with the highest academic average in combined major program
- NSERC Undergraduate Student Research Award** 2017
 Canadian federal fellowships to support academic research as an undergraduate student
- UBC Science Scholar / Dean's Honour List** 2017, 2019
 Awarded to full-time students with winter session GPA above 90%
- Rice University Undergraduate Research Assistantship** 2016
 Rice University's research award to support students pursuing research projects outside their core curriculum
- Rice University President's Honor Roll** 2015
 Awarded to students in top 25% GPA percentile (above 3.882)

Teaching Experience

Lecturer: CS-E400211 “Research Seminar in Embodied Intelligence” - Aalto University 2024

Research seminar for graduate students on intelligent systems with movement capabilities. Master’s-level independent literature review and research course.

Tutor: CS-E4000/E4875 “Research Seminar in Computer Science” - Aalto University 2022-2024

Student supervision via weekly project meetings and personalized lectures. Master’s-level independent literature review and research course. 6 students conferred. Sessions taught: 2022, 2023, 2024.

Co-Instructor: DOM-E5129 “Intelligent Computational Media” - Aalto University 2022,2024

Lectures, assignments, student supervision. Graduate-level applied AI course for art and games. Sessions taught: 2022T3, 2024T3.

Lecturer: CPSC 340 “Machine Learning and Data Mining” - UBC 2021

Sessional lectureship, teaching UBC’s main undergraduate-level machine learning course. 3rd-year level, 160 students, 10 TAs. Sessions taught: 2021S.

TA: CPSC 440/540 “Advanced Machine Learning” - UBC 2021

Teaching Assistant for the graduate level machine learning course, taught by Prof. Mark Schmidt. Sessions taught: 2020W2.

TA: CPSC 330 “Applied Machine Learning” - UBC 2020

Teaching Assistant for the pilot session of the applied machine learning course, taught by Prof. Mike Gelbart. Sessions taught: 2019W2.

TA: CPSC 340 “Machine Learning and Data Mining” - UBC 2018-2019

Teaching Assistant for UBC’s main undergraduate-level machine learning course, taught by Prof. Mark Schmidt, Prof. Frank Wood, and Prof. Mike Gelbart. Responsible for running tutorials, holding office hours, and grading assignments/exams. Sessions taught: 2018W1, 2018W2, 2019W1

Positions & Appointments Held

Research Group Leader - Embodied Intelligence *Helsinki, Finland*

AALTO UNIVERSITY

Sep. 2023 - Dec. 2024

- Founded Aalto MIME Lab - Machine Intelligence for Motion and Embodied Applications
- Lead and organize weekly meetings, provide hands-on project supervision and mentorship, recruit thesis and seminar students
- Supervised 3 MSc theses, 1 BSc thesis.

R&D Scientist Intern - Character Animation • Foundation Models *Helsinki, Finland*

UBISOFT LA FORGE / REDLYNX

Feb. 2023 - Feb. 2024

- Developed reinforcement learning algorithms to enable a rapid motion prototyping system for non-humanoid 3D characters
- Investigated general-purpose use of LoRA-adapted LLMs for goal-driven character control

Lecturer - Computer Science *Vancouver, Canada*

UNIVERSITY OF BRITISH COLUMBIA

May 2021 - Jun. 2021

- CAD 15,000 sessional lectureship for CPSC 340: Machine Learning and Data Mining for the 2021S term.
- Carried out teaching duties in full, including lectures, assignments, exams, and administration.

Applied Scientist Intern - Machine Learning

AMAZON WEB SERVICES, AMAZON.COM, INC.

Vancouver, Canada

May 2020 - Aug. 2020

- Implemented and benchmarked active learning workflow for pedestrian detection by integrating OpenMMLab into AWS SageMaker

Software Development Engineer Intern

AMAZON WEB SERVICES, AMAZON.COM, INC.

Vancouver, Canada

May 2018 - Aug. 2018

- Optimized RDS Oracle's point-in-time restore feature by leveraging dynamic size batch submission algorithm
- Achieved six-fold increase in average throughput from improved feature
- Received full-time return offer

Research Assistant - Bioinformatics

UCANADA CENTRE FOR MOLECULAR MEDICINE AND THERAPEUTICS

Vancouver, Canada

Sep. 2017 - Apr. 2018

- Mentored by Prof. Sara Mostafavi and Dr. Bernard Ng. Funded by NSERC Undergraduate Research Award
- Designed parametric testing methods for non-Euclidean statistical models associating genotypes with brain co-activations
- Implemented large-scale data preprocessing and imputation pipeline involving denoising, alignment, distance mapping, and compression of genetic data

Software Development Engineer Intern

DELIVERY EXPERIENCE, AMAZON.COM, INC.

Vancouver, Canada

Jun. 2017 - Aug. 2017

- Designed and implemented internal editing tool and abstraction logic for customizing and localizing Delivery Benefit Programs in accordance with company directives
- Completed required deliverables and bonus features and deployed product to Amazon's official development platform
- Received return offer for May 2018

Research Assistant - Bioinformatics

BAYLOR COLLEGE OF MEDICINE CENTER FOR GENOME ARCHITECTURE

Houston, United States

Jan. 2016 - Apr. 2016

- Mentored by Prof. Erez Lieberman Aiden
- Developed visualization components of DNA sequencing quality control pipeline to detect physical artifacts in Illumina sequencing experiments

Research Assistant - Human-Computer Interaction

RICE UNIVERSITY COMPUTER SECURITY LAB

Houston, United States

Jan. 2016 - Apr. 2016

- Mentored by Prof. Dan Wallach and Prof. Philip Kortum. Stipend-based Research Assistantship
- Designed, built, and tested the second generation model of STAR-Vote involving sheet-feeding mechanism, audio alert system, barcode detection, and graphical user interface

Frontend Developer Intern

MEDICAL INFORMATICS CORP

Houston, United States

May. 2013 - Aug. 2013

- Medical application / healthcare data management software startup
- Built JavaScript product prototype for live patient monitoring modules for product pitch in startup partnership and sponsorship meetings
- Received return offer

Additional Positions and Activities

Technical Paper Reviewer

SIGGRAPH, EUROGRAPHICS, SPRINGER NATURE, IEEE TVCG, ICCV

2022 - PRESENT

- Technical paper program reviewer in character animation and motion control

Music Director • Peer Advisor*Vancouver, Canada*

TAPESTRY CHURCH AT MARPOLE

2017 - 2020

- Ministerial work involving coordination for weekly gathering, counseling, music practice and performance

Athlete • Competition Announcer*Vancouver, Canada*

UCANADA WEIGHTLIFTING AND POWERLIFTING CLUB

2016 - 2019

- In association with British Columbia Powerlifting Association (CanadaPA)

Conference Staff*Seoul, Republic of Korea*

IEEE INTELLIGENT VEHICLES SYMPOSIUM

2015

- Facilitated 5-day conference providing interpretation and technical support

Translator • Squad Leader*Seoul, Republic of Korea*

REPUBLIC OF KOREA ARMY

2013 - 2015

- English-Korean translation and interpretation duties involving communications and computing devices for the United States Forces Korea.

Actor • Ensemble Member*Houston, United States*

RICE UNIVERSITY MUSICAL THEATRE

2012, 2013, 2015

- Participated in the production of: The Drowsy Chaperone (2012), The Producers (2013), and a theatrical adoption of Black Mirror (2015)